Algebra 2 Review: Functions & Transformations

Station 1 of 4: Function Notation, Relations, Domain, and Range

Name: _____ Date: _____

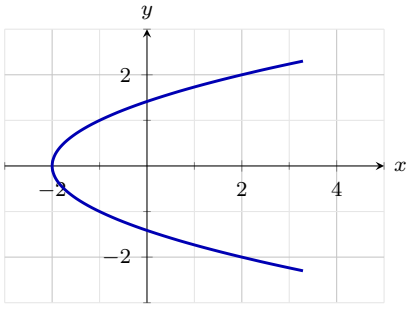
Give domains and ranges in interval notation unless told otherwise. Show how you decide.

1. Let $f(x) = 2x^2 - 5x + 1$. Find $f(-3)$.

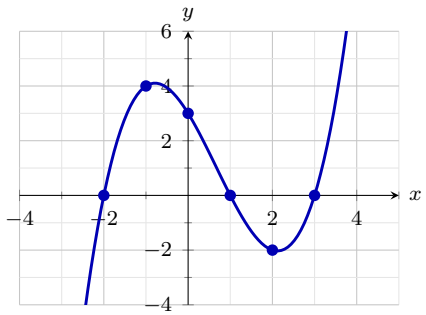
2. Let $f(x) = x^2 - 4x$. Simplify $f(a + 2)$.

3. Is each relation a function? If so, state the domain and range. $A = \{(1, 3), (2, 5), (3, 3), (4, 7)\}$ and $B = \{(2, -1), (5, 0), (2, 3)\}$.

4. Does the graph show y as a function of x? Use the vertical line test and name two points that prove your answer.



5. The graph of f is shown. Find $f(-1)$, $f(2)$, and all x with $f(x) = 0$.



6. State the domain of $f(x) = \sqrt{5 - 2x}$ in interval notation.

7. State the domain of $g(x) = \dfrac{x + 4}{x^2 - 9}$ in interval notation.

8. State the domain and range of $h(x) = -|x - 2| + 3$.

9. A truck rental costs $C(m) = 35 + 0.25m$ dollars for m miles. Find $C(120)$ and solve $C(m) = 80$. What does each answer mean?

10. State the domain of $f(x) = \dfrac{\sqrt{x + 3}}{x - 1}$ in interval notation.

Algebra 2 Review: Functions & Transformations

Station 2 of 4: Parent Functions and Transformations of Graphs

Name: _____ Date: _____

Apply transformations in the correct order. Remember that $f(x+3)$ moves the graph LEFT.

1. Describe, in order, how $g(x) = -2|x+3|+1$ is obtained from $y = |x|$. State the vertex.

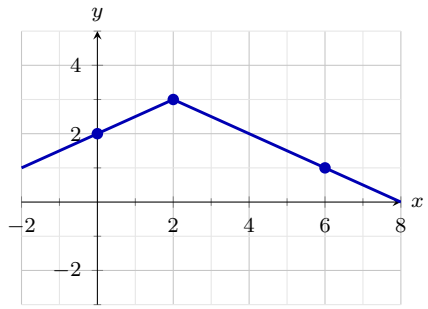
2. Describe how $h(x) = -\frac{1}{2}(x-4)^2 + 2$ is obtained from $y = x^2$. State the vertex and the range.

3. The point $(-2, 5)$ is on $y = f(x)$. Find its image on $y = f(x+3) - 4$.

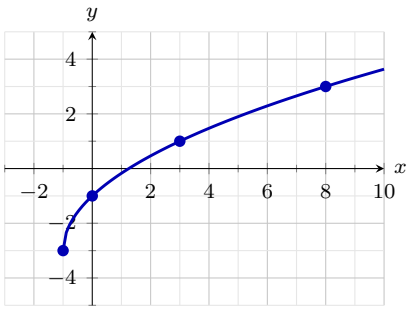
4. The point $(6, -2)$ is on $y = f(x)$. Find its image on $y = -f\left(\frac{1}{2}x\right)$.

5. Write the equation for $y = \sqrt{x}$ reflected in the y-axis and then shifted down 1. State the domain and range.

6. Write an equation for the graph, which is a transformation of $y = |x|$.



7. Write an equation for the graph, which is a transformation of $y = \sqrt{x}$.



8. Are the graphs of $y = 2x^2 + 3$ and $y = 2(x^2 + 3)$ the same? Give the vertex of each and explain how order matters.

9. Describe $g(x) = \left(\frac{1}{2}x\right)^3 - 2$ as a transformation of $y = x^3$, then find $g(4)$.

10. The graph of $y = \frac{1}{x}$ is shifted right 2 and up 5. Write the equation and give both asymptotes.

Algebra 2 Review: Functions & Transformations

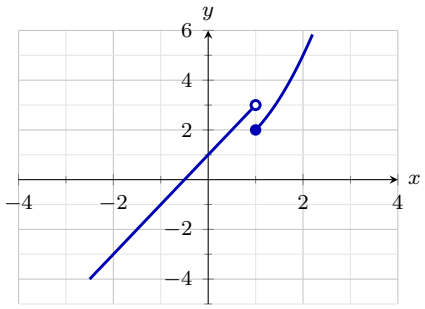
Station 3 of 4: Piecewise Functions, Operations, and Composition

Name: _____ Date: _____

Read $(f \circ g)(x)$ as $f(g(x))$: work from the inside out.

1. Let $f(x) = \begin{cases} x+4, & x < -1 \\ x^2, & -1 \le x \le 2 \\ 6-x, & x > 2 \end{cases}$. Find $f(-3)$, $f(-1)$, $f(2)$, and $f(3)$.

2. The graph of a piecewise function f is shown. Find $f(-1)$, $f(1)$, and $f(2)$. Is f continuous? Explain.



3. Rewrite $g(x) = |x-3| + 1$ as a piecewise-defined function.

4. Let $f(x) = x^2 - 1$ and $g(x) = 3x + 2$. Find $(f+g)(x)$ and $(f \cdot g)(2)$.

5. Let $f(x) = x^2 + 2x$ and $g(x) = x - 4$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

6. Let $f(x) = \sqrt{x}$ and $g(x) = x^2 - 9$. Find $(f \circ g)(x)$ and its domain.

7. Use the table to find $(f \circ g)(3)$, $(g \circ f)(1)$, and $(f \circ f)(2)$.

x	1	2	3	4
$f(x)$	4	1	2	3
$g(x)$	3	4	1	2

8. A circular ripple has radius $r(t) = 3t$ cm after t seconds, and its area is $A(r) = \pi r^2$. Write $A(r(t))$ and find the area at $t = 4$ seconds.

9. Temperature in Celsius is $C(F) = \frac{5}{9}(F - 32)$ and in kelvins is $K(C) = C + 273.15$. Write $K(C(F))$ and find $K(C(212))$.

10. Write $h(x) = (5x-2)^4$ as a composition $h(x) = f(g(x))$ of two simpler functions.

Algebra 2 Review: Functions & Transformations

Station 4 of 4: Inverse Functions

Name: _____ Date: _____

Remember that $f^{-1}(x)$ *means the inverse of* f*, not* $\dfrac{1}{f(x)}$.

1. Find the inverse of $f(x) = 3x - 7$.

2. Find the inverse of $f(x) = \dfrac{2x+1}{x-3}$.

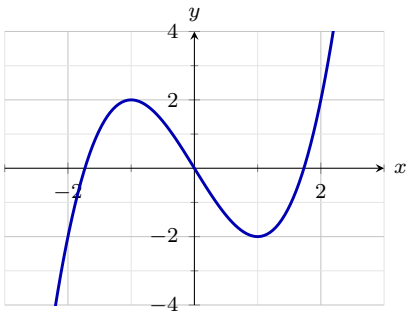
3. Find the inverse of $f(x) = (x-2)^2 + 1$ for $x \geq 2$, and state its domain.

4. Verify by composition that $f(x) = x^3 + 1$ and $g(x) = \sqrt[3]{x-1}$ are inverses.

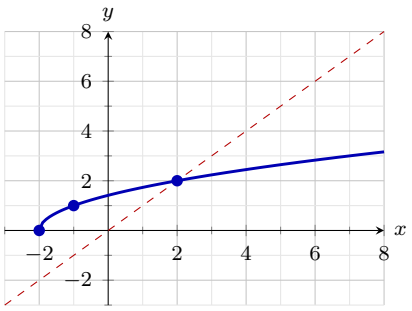
5. The function f is given by the table. Find $f^{-1}(7)$, $f^{-1}(-1)$, and $\left(f^{-1} \circ f\right)(4)$.

x	0	1	2	4
$f(x)$	3	7	-1	5

6. The graph of $f(x) = x^3 - 3x$ is shown. Is f one-to-one? Use the horizontal line test and name a line that proves it.



7. The graph of $f(x) = \sqrt{x+2}$ and the line $y = x$ are shown. Give the points on f^{-1} that correspond to the three marked points, then write $f^{-1}(x)$ and its domain.



8. A taxi charges $c(m) = 2.50 + 1.60m$ dollars for m miles. Find $c^{-1}(x)$ and use it to find how far you can ride for \$18.50.

9. A student says the inverse of $f(x) = 2x + 6$ is $\dfrac{1}{2x+6}$. Explain the error, find the correct inverse, and compare the two at $x = 4$.

10. State the domain and range of $f(x) = \sqrt{x-1} + 3$ and of f^{-1}.

Algebra 2 Review: Functions & Transformations

Answer Key

Station 1: Function Notation, Relations, Domain, and Range

1. 34
   $2(9) + 15 + 1$
2. $a^2 - 4$
   $(a+2)^2 - 4(a+2) = a^2 + 4a + 4 - 4a - 8$
3. A is a function: domain $\{1, 2, 3, 4\}$, range $\{3, 5, 7\}$. B is not: $x = 2$ gives both -1 and 3.
4. No: the vertical line $x = 2$ meets the graph at $(2, 2)$ and $(2, -2)$
   graph of $x = y^2 - 2$
5. $f(-1) = 4$, $f(2) = -2$; $x = -2,\ 1,\ 3$
   zeros are the x-intercepts
6. $\left(-\infty, \frac{5}{2}\right]$
   $5 - 2x \geq 0 \Rightarrow x \leq \frac{5}{2}$
7. $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$
   $x^2 - 9 \neq 0$*; the numerator may be* 0
8. Domain: all reals; range: $(-\infty, 3]$
   vertex $(2, 3)$*, opens down*
9. $C(120) = \$65$: 120 miles cost \$65. $m = 180$: \$80 pays for 180 miles.
10. $[-3, 1) \cup (1, \infty)$
    $x + 3 \geq 0$ *and* $x \neq 1$

Station 2: Parent Functions and Transformations of Graphs

1. Left 3, vertical stretch by 2, reflect in the x-axis, up 1; vertex $(-3, 1)$
2. Right 4, vertical compression by $\frac{1}{2}$, reflect in the x-axis, up 2; vertex $(4, 2)$; range $(-\infty, 2]$
3. $(-5, 1)$
   x *shifts LEFT 3:* $-2 - 3 = -5$*; y:* $5 - 4 = 1$
4. $(12, 2)$
   horizontal stretch by 2: $6 \cdot 2 = 12$*; reflect:* $-(-2) = 2$
5. $y = \sqrt{-x} - 1$; domain $(-\infty, 0]$; range $[-1, \infty)$
6. $y = -\frac{1}{2}|x - 2| + 3$
   vertex $(2, 3)$*; slope* $-\frac{1}{2}$ *on the right*
7. $y = 2\sqrt{x+1} - 3$
   starts at $(-1, -3)$*; passes through* $(3, 1)$
8. No: vertices $(0, 3)$ and $(0, 6)$
   stretch by 2 first and then shift up 3, versus shift up 3 first and then stretch
9. Horizontal stretch by 2, down 2; $g(4) = 6$
   $\left(\frac{1}{2} \cdot 4\right)^3 - 2 = 8 - 2$
10. $y = \frac{1}{x-2} + 5$; vertical asymptote $x = 2$, horizontal asymptote $y = 5$

Station 3: Piecewise Functions, Operations, and Composition

1. $f(-3) = 1$, $f(-1) = 1$, $f(2) = 4$, $f(3) = 3$
   check which inequality each input satisfies
2. $f(-1) = -1$, $f(1) = 2$, $f(2) = 5$; not continuous: it jumps at $x = 1$ (open circle $(1, 3)$, closed $(1, 2)$)
3. $g(x) = -x + 4$ for $x < 3$; $g(x) = x - 2$ for $x \geq 3$
   $|x - 3| = 3 - x$ *when* $x < 3$
4. $(f + g)(x) = x^2 + 3x + 1$; $(f \cdot g)(2) = 24$
   $f(2) = 3$ *and* $g(2) = 8$
5. $(f \circ g)(x) = x^2 - 6x + 8$; $(g \circ f)(x) = x^2 + 2x - 4$
   $f(g(x)) = (x-4)^2 + 2(x-4)$
6. $\sqrt{x^2 - 9}$; domain $(-\infty, -3] \cup [3, \infty)$
   need $x^2 - 9 \geq 0$
7. 4, 2, 4
   $f(g(3)) = f(1)$*;* $g(f(1)) = g(4)$*;* $f(f(2)) = f(1)$
8. $A(r(t)) = 9\pi t^2$; area $= 144\pi$ cm^2
   $r(4) = 12$ *and* $\pi(12)^2$
9. $K(C(F)) = \frac{5}{9}(F - 32) + 273.15$; $K(C(212)) = 373.15$ kelvins
   water boils at $212°F$
10. $f(x) = x^4$ and $g(x) = 5x - 2$
    inside function first (answers may vary)

Station 4: Inverse Functions

1. $f^{-1}(x) = \frac{x+7}{3}$
   swap x *and* y*, then solve for* y
2. $f^{-1}(x) = \frac{3x+1}{x-2}$
   $xy - 3x = 2y + 1 \Rightarrow y(x-2) = 3x + 1$
3. $f^{-1}(x) = \sqrt{x-1} + 2$; domain $x \geq 1$
   range of f *is* $y \geq 1$
4. $f(g(x)) = (\sqrt[3]{x-1})^3 + 1 = x$ and $g(f(x)) = \sqrt[3]{x^3} = x$
5. 1, 2, 4
   swap inputs and outputs
6. No: any line $y = k$ with $-2 < k < 2$ meets it 3 times; for example $y = 0$ at $x = -\sqrt{3},\ 0,\ \sqrt{3}$
7. $(0, -2)$, $(1, -1)$, $(2, 2)$; $f^{-1}(x) = x^2 - 2$, domain $x \geq 0$
   reflect in $y = x$*: swap coordinates*
8. $c^{-1}(x) = \frac{x - 2.50}{1.60}$; $c^{-1}(18.50) = 10$ miles
   the inverse takes a cost and returns miles
9. f^{-1} means the inverse, not the reciprocal. $f^{-1}(x) = \frac{x-6}{2}$, so $f^{-1}(4) = -1$, but $\frac{1}{f(4)} = \frac{1}{14}$
10. f: domain $[1, \infty)$, range $[3, \infty)$; f^{-1}: domain $[3, \infty)$, range $[1, \infty)$
    domain and range swap